

QALHAT, Oman

WMO: 412670

Lat: 22.667N

Lon: 59.400E

Elev: 36

StdP: 14.68

Time Zone: 4.00 (E04)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	65.2	66.6	34.3	29.1	83.3	38.8	34.8	80.9	45.9	73.3	37.3	72.7	6.2	280	0.403

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	9.5	105.2	73.7	102.3	74.2	100.1	74.8	86.1	92.7	85.1	92.2	84.1	91.6	11.8	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
84.6	181.7	90.8	83.3	174.2	90.4	82.1	167.5	90.2	51.1	92.6	49.8	92.4	48.6	91.8	95.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.0	21.8	18.1	62.3	111.9	1.4	3.6	61.3	114.5	60.4	116.6	59.6	118.6	58.6	121.1		
(5)			49.1	86.8	1.9	1.7	47.7	88.0	46.6	89.0	45.6	89.9	44.2	91.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	83.6	71.8	73.4	78.3	85.2	93.2	93.4	91.5	88.7	88.3	85.4	79.4	74.3		
	DBStd	8.30	2.20	3.48	3.90	4.29	4.43	4.35	4.60	3.89	2.66	2.74	2.44	2.29		
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD65	1	0	1	0	0	0	0	0	0	0	0	0	0		
	CDD50	12272	677	655	877	1056	1338	1301	1287	1198	1148	1099	883	753		
	CDD65	6797	212	235	412	606	873	851	822	733	698	634	433	288		
	CDH74	87770	440	991	3470	7969	14134	13707	12681	10618	10040	8471	4033	1216		
	CDH80	49046	12	124	843	3874	9671	9388	8221	6165	5723	4094	878	53		
(14) Wind	WSAvg	7.0	9.0	9.1	7.5	6.6	6.4	6.5	6.0	5.7	5.2	5.7	7.6	9.1		
(15) Precipitation	PrecAvg	4.1	0.3	0.4	0.7	0.7	0.2	0.7	0.2	0.1	0.0	0.1	0.2	0.4		
	PrecMax	10.7	3.4	3.6	4.3	2.1	1.8	8.8	1.5	0.6	0.2	1.1	3.0	3.3		
	PrecMin	0.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
	PrecStd	2.6	0.7	0.7	1.1	0.7	0.4	1.9	0.4	0.1	0.0	0.3	0.7	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	81.5	87.9	94.9	100.2	108.0	108.6	110.0	104.8	100.1	96.2	89.9	83.5		
		MCWB	66.5	63.8	66.4	67.9	72.7	72.6	73.7	75.0	72.4	70.2	67.5	67.3		
	2%	DB	78.9	83.4	89.1	96.8	104.0	104.4	104.1	98.9	95.8	93.2	86.9	81.3		
		MCWB	67.2	67.0	67.4	68.6	72.9	73.1	76.9	77.5	74.9	73.0	70.9	68.3		
	5%	DB	77.7	80.6	86.3	93.8	101.6	101.7	99.9	95.8	93.7	91.5	85.4	79.9		
		MCWB	66.8	67.9	67.9	70.1	73.1	74.9	77.9	78.2	76.5	74.7	71.6	68.2		
	10%	DB	76.4	78.8	84.2	91.7	99.7	99.4	97.3	93.5	92.2	90.0	84.1	78.8		
		MCWB	66.1	67.6	69.0	71.2	73.4	76.5	79.4	79.1	77.6	74.4	71.4	67.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	71.4	73.8	76.8	80.3	85.4	87.0	89.1	86.0	85.3	81.8	77.2	72.7		
		MCDB	77.9	79.0	83.3	87.5	94.7	92.8	94.0	92.2	89.7	88.3	84.1	79.5		
	2%	WB	69.9	72.3	75.2	79.0	83.7	85.5	86.6	84.6	83.6	80.3	75.7	71.2		
		MCDB	76.7	78.3	81.9	86.7	93.2	93.0	93.1	90.9	89.6	88.2	83.7	78.6		
	5%	WB	68.6	70.9	74.0	77.9	82.5	84.4	85.4	83.6	82.6	79.1	74.4	70.1		
		MCDB	76.0	77.5	81.0	86.2	92.2	92.9	92.2	90.1	89.3	87.7	83.0	78.2		
	10%	WB	67.4	69.5	72.8	76.8	81.1	83.4	84.4	82.7	81.6	78.0	73.3	69.0		
		MCDB	75.2	76.7	80.4	85.8	92.1	92.4	91.6	89.4	88.9	87.3	82.3	77.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.9	8.5	8.8	8.7	9.5	8.6	7.6	7.1	7.1	8.3	8.2	7.7		
		MCDBR	10.0	10.9	12.2	12.1	12.8	13.0	12.9	11.5	9.7	9.6	9.5	9.0		
	5% WB	MCWBR	8.0	9.5	11.1	11.7	11.3	11.8	10.6	9.9	11.1	12.8	11.3	7.6		
		MCWBR	8.5	8.3	8.3	8.8	9.7	8.4	7.9	7.6	6.8	7.6	8.1	8.1		
(40) Clear-Sky Solar Irradiance	taub	taub	0.445	0.493	0.563	0.632	0.680	0.764	0.835	0.775	0.649	0.509	0.469	0.443		
		taud	2.119	1.970	1.793	1.683	1.606	1.488	1.423	1.496	1.666	1.930	2.056	2.119		
	Edn at Noon	Edn at Noon	257	252	240	226	214	195	182	193	216	243	246	251		
		Edh at Noon	45	56	70	79	85	95	101	94	78	57	48	44		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1415	1672	1930	2119	2267	2139	1977	1980	1928	1769	1483	1344		
	RadStd	RadStd	59	63	90	69	35	89	101	51	35	44	51	57		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (7 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		+0.41	N/A	N/A	N/A	+0.70	+0.81	N/A	N/A	+173
										+176

Nomenclature: See separate page